The macroeconomic landscape

Jean-Félix Brouillette

Session 1

HEC Montréal | MBA

What is macroeconomics?

Macroeconomics deals with…

- Global economic phenomena: the **big picture**
- **Interconnections** between economic agents (e.g., consumers, firms, gov.)
- **Long-run** trends, **short-run** fluctuations, and economic **policy**

Macroeconomics is closely linked to many other fields:

- Politics and geopolitics
- Psychology and sociology
- Finance
- Environment, technology, etc.

Macroeconomics vs. microeconomics

Microeconomics provides great tools to study how individuals make decisions...

...but not how they **aggregate** to the economy as a whole

...and this aggregation, in turn, matters **a lot** for individual decision-making!

When individual rationality leads to collective madness...

- **The paradox of thrift:** everyone saves more in a recession
- **Helicopter money:** too many dollars chasing too few goods
- **The fallacy of composition:** standing up at a hockey game

Why should *you* care about macroeconomics?

The success of any business decision rests partly on **macroeconomic conditions**

Macroeconomic events, forces, and trends…

1. …disrupt **business strategies**
2. …influence **costs, wages, interest rates**, and **competition**
3. …are sources of **risk**, but also of **opportunity**
4. …shape **markets** on a global scale

…and **macroeconomic jargon** will be at the heart of conversations you will have as future managers

A complex and uncertain world

E "Who wrangled the best trade deal from Donald Trump?"

FT "Gold rips past $5,000 on fears of global turmoil"

THE GLOBE AND MAIL "Mark Carney's speech got the world's attention—but was it the right message?"

B "Kevin Warsh's Fed Job Already Looks Impossible"

LA PRESSE "Les exportations du Québec en forte baisse"

WSJ "The Big Money in Today's Economy Is Going to Capital, Not Labor"

A complex and uncertain world

“Who wrangled the best trade deal from Donald Trump?”

“Kevin Warsh’s Fed Job Already Looks Impossible”

“Gold rips past $5,000 on [illegible]”

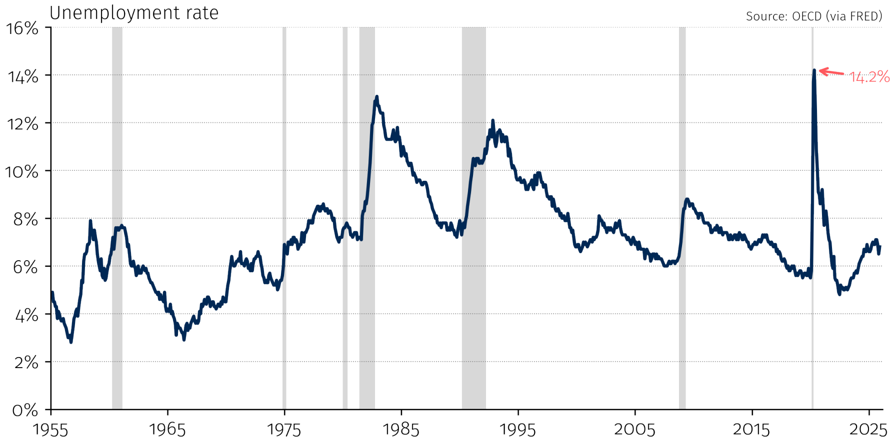
“Les exportations d[illegible] Québec [illegible] baisse”

Where are we now?

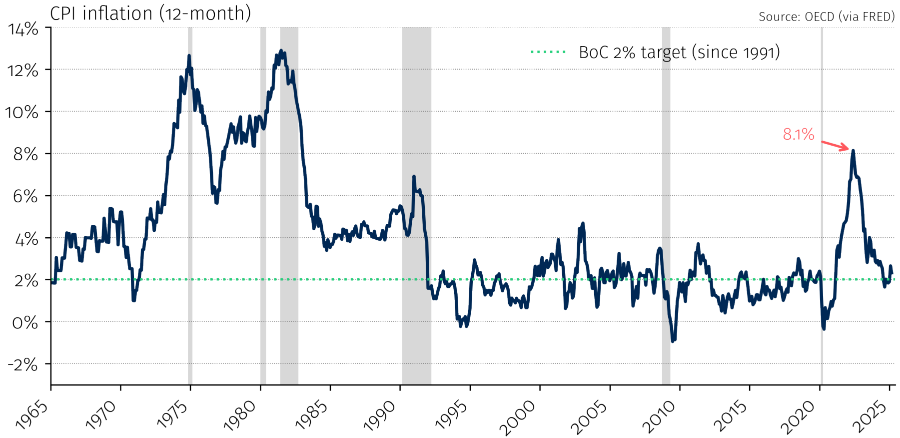
“Mark Carney’s speech got the world’s attention—but was it the right message?”

“The Big Money in Today’s Economy Is Going to Capital, Not Labor”

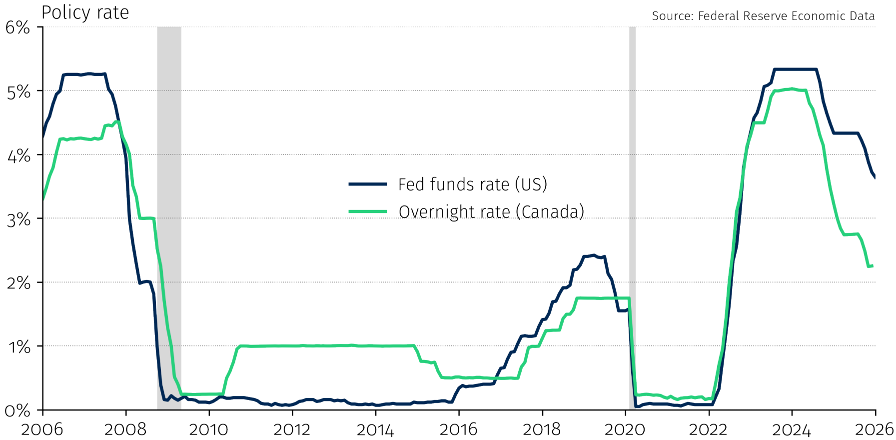
After a very unusual recession…



An old enemy back from the dead...

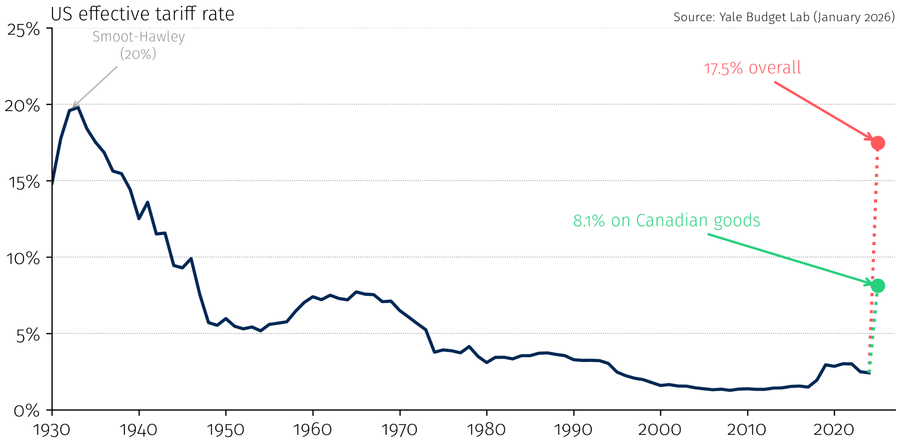


A forceful response…

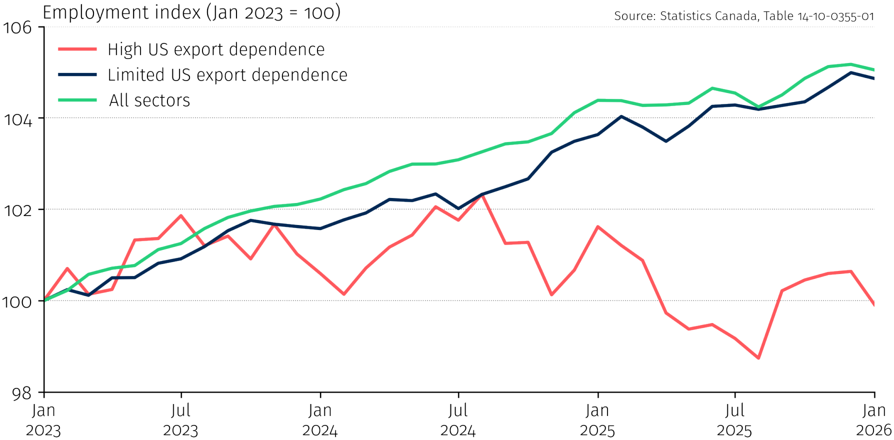


A return to normal…

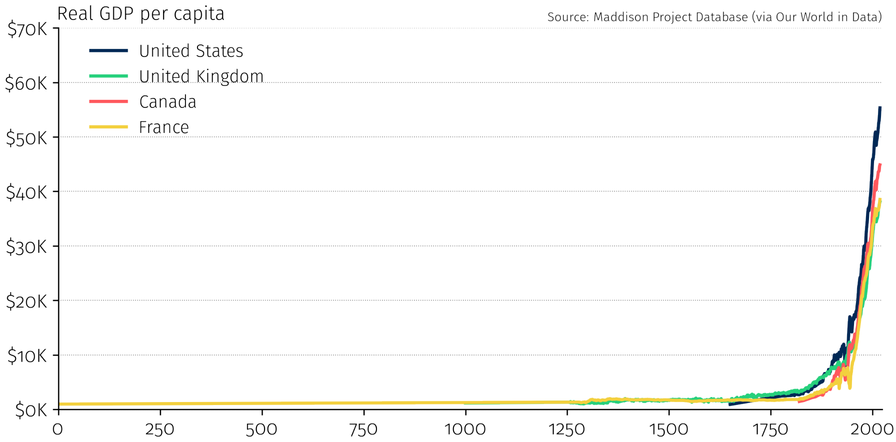
The return of protectionism…



An uncertain future…



What about the distant future?



Outline

1. **Course overview and logistics**
2. Measuring the economy: GDP
3. Comparing GDP over time
4. Comparing GDP across countries
5. Beyond GDP

Course overview

Course objectives

No, I will not make you economists...but by the end of this course, you will:

1. Be **less intimidated** by macroeconomic discussions and jargon
2. Be able to critically interpret **macroeconomic news** (“BS” detector)
3. Be aware of major **macroeconomic trends**
4. Have **objective and coherent** conversations about macroeconomics

What this course is not

Technical. Abstract. Ideological.

Themes to be covered

1. **Long-run economic growth:** why are some countries rich and others poor?
2. **The labor market:** inequality, technology, and AI
3. **Business cycles:** why does the economy boom and bust?
4. **Monetary and fiscal policy:** how governments fight recessions
5. **International trade and imbalances:** (de)globalization, tariffs, etc.

Teaching team

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- Office: 4.148 (CSC)

Pedagogical material

ZoneCours (ZC)

- Slides and articles posted before each class
- Exercises and solutions

Textbook

- Vincent & Yared (Pearson Revel): not mandatory
- ZC: reference for lectures + exercises

Stay on top of macro news

- *The Economist, The Financial Times, The Wall Street Journal, Bloomberg, La Presse Affaires, Reuters, Les Échos, etc.*

Grades

Team project (two parts)	**30%**
Quiz (session 4, ~15 minutes, multiple choice)	**10%**
Final exam (March 30th)	**50%**
Contribution and professionalism	**10%**

Details

Team project details are on ZC. Examples of quiz questions will be provided.

Contribution and professionalism

Criteria:

1. **Attendance and punctuality** — as per program requirements
2. **Active participation** — quality and relevance of interventions
3. **Professionalism** — autonomy, judgment, collegiality

To promote active learning:

1. **Name cards** — please bring yours to every class
2. **No laptops** in class (exception: tablets lying flat)
3. **Limit comings and goings** during class

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Measuring the economy: GDP

What is GDP?

Gross Domestic Product (GDP)

The total **market value** of all **final** goods and services produced **within a country** during a **given period**.

Key words:

- **Market value:** measured in dollars (or another currency)
- **Final:** excludes intermediate goods to avoid double-counting
- **Within a country:** production on Canadian soil
- **Given period:** a flow, not a stock (typically quarterly or annual)

Three ways to measure GDP

They all give the same answer

1. Expenditure

Who is buying?

$$Y = C + I + G + NX$$

- C: consumption
- I: investment
- G: gov. spending
- NX: net exports

2. Income

Who is earning?

- Wages/salaries
- Corporate profits
- Rental income
- Interest income

3. Value added

Who is producing?

- Value added at each stage
- Revenue − int. inputs

Key insight

Every dollar spent is a dollar earned is a dollar of value created

Example: value added avoids double counting

Example

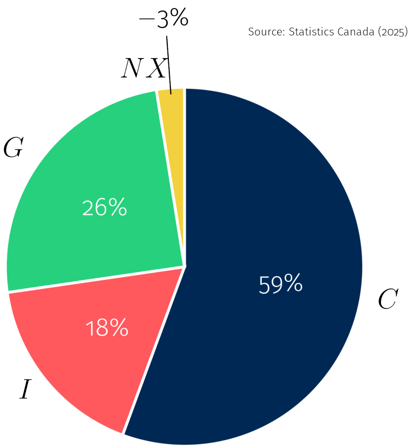
A business hires employees to work in a fertilizer plant and pays them \$0.50 in wages. The business sells the fertilizer to Dwight, a self-employed farmer, for \$0.80. Dwight uses it to produce beets, which he sells to Jim for \$1. Jim eats the beets.

Production		**Income**		**Expenditure**	
Manufacturing	0.80	Wages	0.50	Consumption	1.00
Agriculture	0.20	Profits	0.30		
		Prop. income	0.20		
Total	**1.00**	**Total**	**1.00**	**Total**	**1.00**

Key insight

Production = **value added** (revenue − intermediate inputs), not total revenue. This avoids double counting and ensures all three approaches agree.

The expenditure approach: $Y = C + I + G + NX$



Consumption (C)

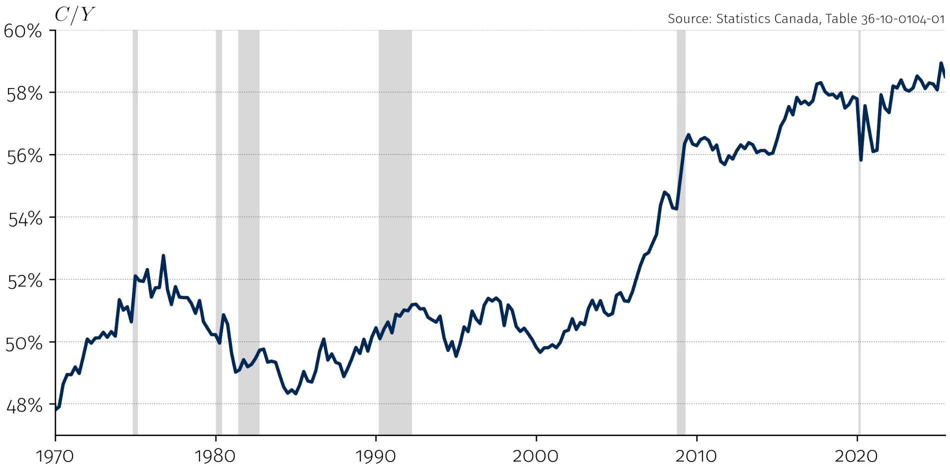
Household spending on final goods and services:

- **Durable goods** — cars, furniture, appliances (long-lasting)
- **Non-durable goods** — food, clothing, gasoline (used up quickly)
- **Services** — rent, healthcare, haircuts, education

Key insight

Services account for over 60% of consumption in Canada. The shift from goods to services is a hallmark of advanced economies.

Consumption share of GDP in Canada



Consumption is the largest component (≈60%), and increasing over time.

Investment (I)

Spending on goods used to produce future output:

- **Business fixed capital** — machines, equipment, factories, software
- **Residential** — new housing construction
- **Inventories** — unsold goods stored in warehouses

Common confusion

In economics, "investment" means **physical capital formation**, not buying stocks or bonds. A firm purchasing a delivery van is investment; buying shares of Tesla is not.

Example: investment

Example

General Motors produces a minivan, which it sells for \$1,000 to the Michael Scott Paper Company. The cost of producing it is made up of workers' wages worth \$600.

Production		**Income**		**Expenditure**	
Manufacturing	1,000	Wages	600	Investment	1,000
		Profits	400		
Total	**1,000**	**Total**	**1,000**	**Total**	**1,000**

Key insight

A firm buying a capital good = **investment**, not consumption.

Example: inventories (Year 1)

Example

Dunder Mifflin produces 500 tons of white paper worth $400 and stores them in its warehouse while it waits for customers to buy them. The cost of producing them is made up of workers' wages of $500.

Production		Income		Expenditure	
Manufacturing	400	Wages	500	Investment	400
		Profits	(100)		
Total	**400**	**Total**	**400**	**Total**	**400**

Key insight

Unsold goods = **inventory investment**. GDP captures all production, even without a sale.

Example: inventories (Year 2)

Example

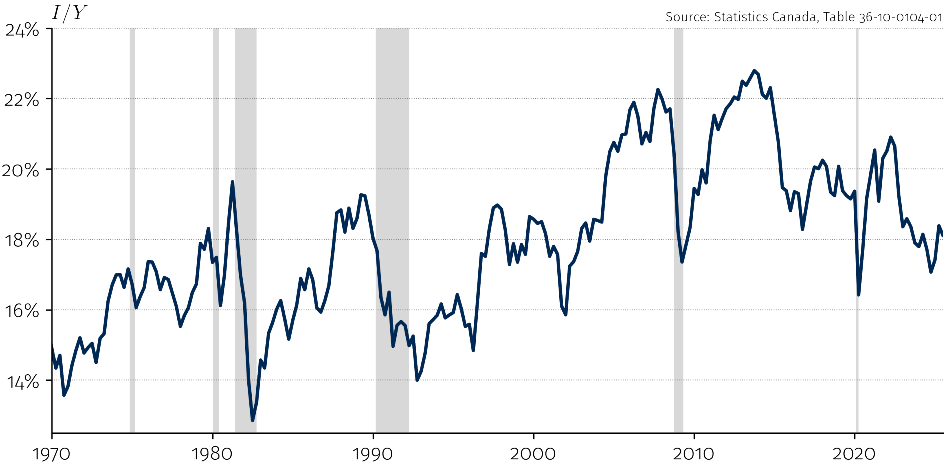
The following year, Dunder Mifflin finally sells its paper worth \$400.

Production		Income		Expenditure	
				Consumption	400
				Investment	(400)
Total	**0**	**Total**	**0**	**Total**	**0**

Key insight

Selling from inventory: C rises, I falls $\Rightarrow$ GDP = 0. Production was counted in year 1.

Investment share of GDP in Canada



Investment is the most **volatile** component — it swings sharply during recessions.

Government spending (G)

Major producer of goods/services: military, education, healthcare, infrastructure.

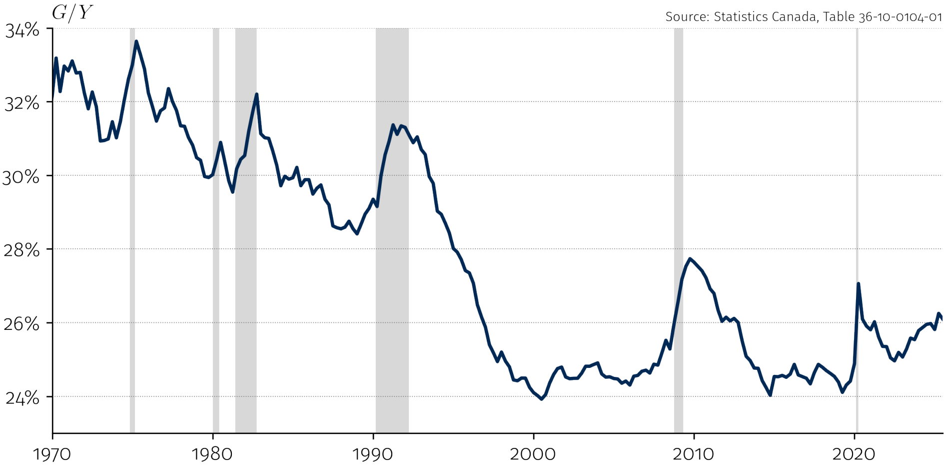
Valued at cost

Often missing market prices, so G measured using **cost of inputs**.

Transfers $\neq$ government spending

Social transfers (UI, pensions, welfare) are **not** in G because they do not involve any new production — they redistribute existing income.

Government spending share of GDP in Canada



Government spending spikes in recessions, but on a long downward trend since the 1990s.

Exports and imports (NX)

C, I, and G include spending on **foreign** goods (imports).

Since GDP measures *domestic* production, we need a correction:

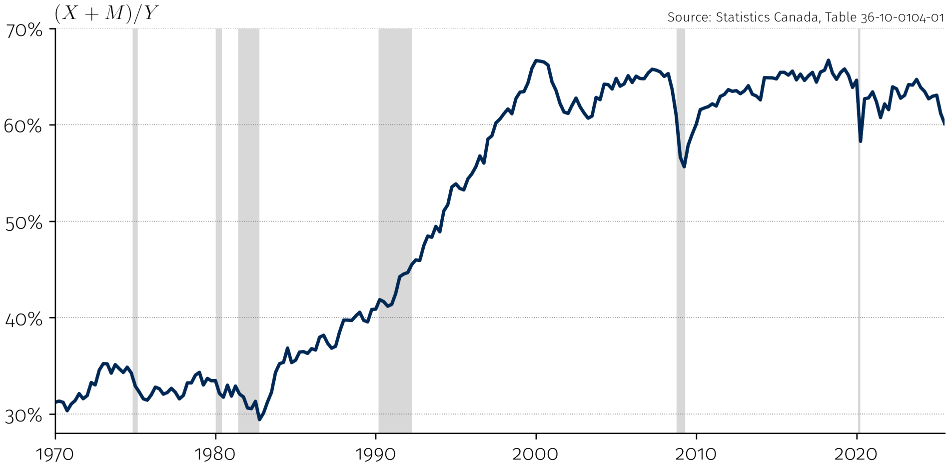
$$Y = \underbrace{C + I + G}_{\text{total spending}} + \underbrace{X - M}_{\text{net exports}}$$

- X (exports): domestic production sold to foreigners
- M (imports): foreign production already counted in C, I, G

Trade openness

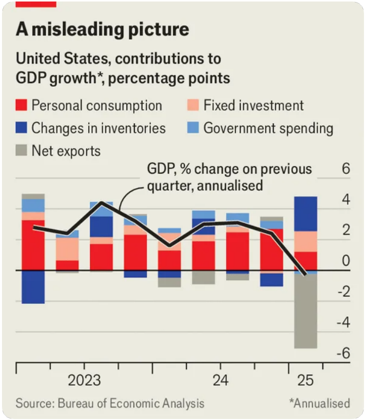
$(X + M)/Y$ measures how important trade is for an economy.

Trade share of GDP in Canada



Canada is a very open economy. Trade peaked with NAFTA (1994) and has slightly declined since.

Training your "BS" detector



Source: *The Economist*, 2025

In Q1 2025, US GDP fell. Many blamed imports: "the trade deficit subtracted 5 pp from GDP growth."

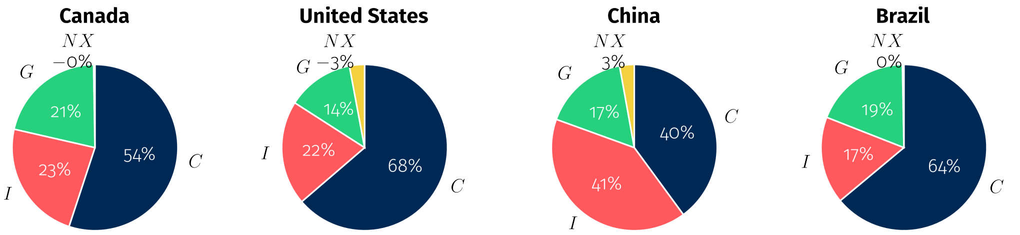
That's misleading

Imports enter GDP **twice**: once *added* in C, I, or G, and once *subtracted* in NX.

If firms front-load imports before tariffs, inventories surge ($\uparrow I$) while net exports plunge ($\downarrow NX$). The two cancel out.

If consumers hurry to buy dishwashers before tariffs, consumption surges ($\uparrow C$) while net exports plunge ($\downarrow NX$). The two cancel out.

GDP by expenditure: cross-country comparison



Source: World Bank (2024)

Different growth models

China is investment-driven (I = 41%), while the US is consumption-driven (C = 68%). These differences shape trade patterns and growth strategies.

What GDP does *not* measure

GDP only counts **market transactions**. A lot of economic activity is missed:

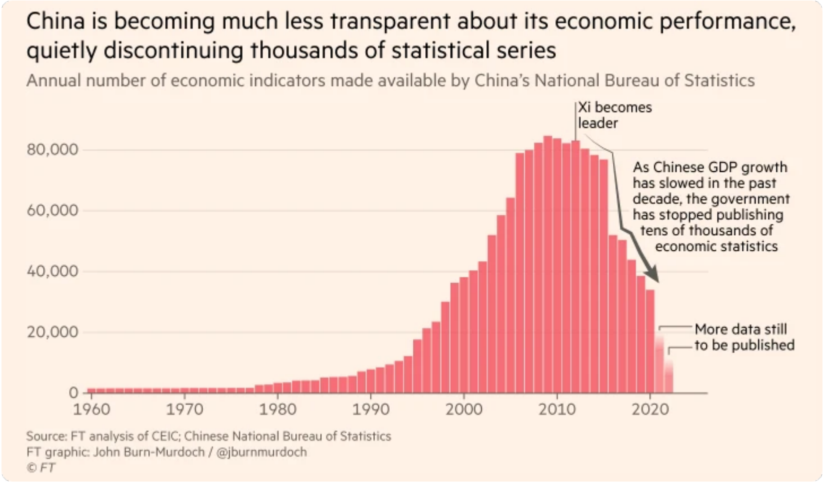
- **Household production** — cooking, childcare, home repairs
- **Volunteer work** — community services, charities
- **Underground economy** — unreported income, informal work

Sex, drugs, and GDP

In 2014, EU accounting rules began requiring countries to include estimates of illegal activities (e.g., drugs, prostitution) in GDP. Italy's GDP rose by 18% overnight...

The Economist, 2014

Can we trust GDP figures?



GDP manipulation: three cautionary tales

- **China:** Suspiciously smooth GDP growth, statistical series are increasingly discontinued, and provinces routinely report figures matching central targets... Autocracies overstate growth by ~35% on average.
  Martinez, "How Much Should We Trust the Dictator's GDP Growth Estimates?" *JPE*, 2022

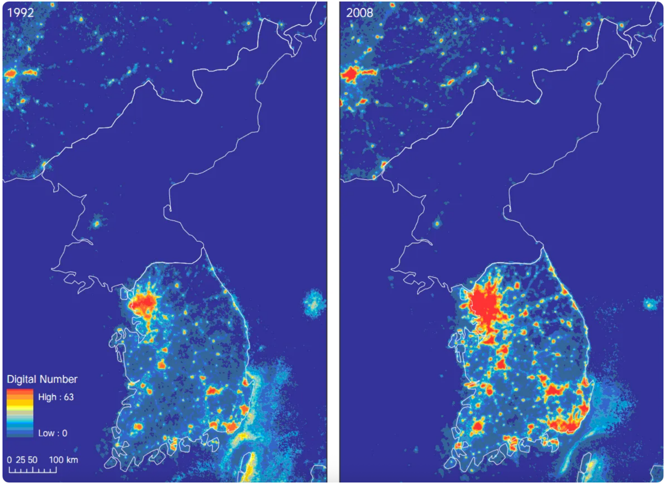
- **Venezuela:** the central bank stopped publishing GDP and inflation data from 2016 to 2019; when it resumed, it revealed a cumulative GDP contraction of ~47%
  IMF World Economic Outlook; Banco Central de Venezuela, 2019

- **Greece:** the 2009 deficit was reported as 3.7% of GDP, then revised to 15.4% — more than four times the original figure, triggering the eurozone debt crisis
  European Commission, Report on Greek Government Deficit and Debt Statistics, 2010

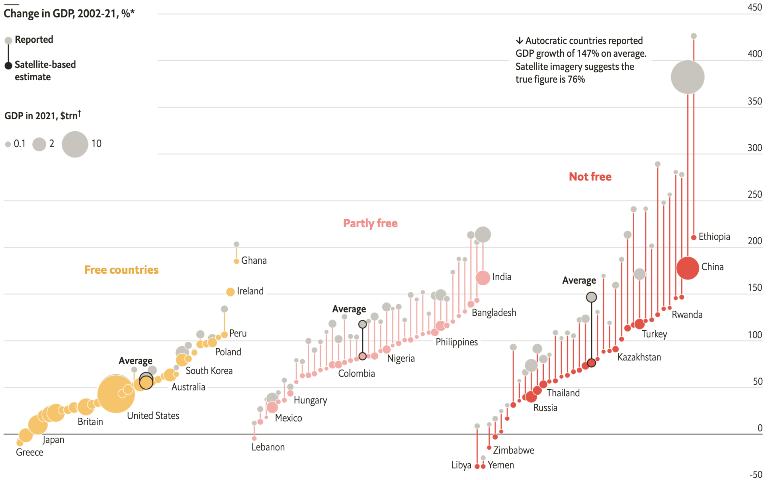
When data goes missing…economists get creative!

- **Satellite night lights:** brightness correlates with economic activity — for poor-data countries, night lights are as informative as official GDP, (Henderson et al., *AER*, 2012).
- **Trade and port data:** shipping volumes, ship AIS draft depth, customs records, and container throughput reveal trade flows that official statistics may undercount.
- **ML inference:** ML algorithms detect oil tankers turning off transponders to evade sanctions (dark shipping), mapping illicit trade flows in real time, (Fernández-Villaverde et al., NBER WP 33486, 2025).
- **Online prices:** web-scraped retail prices track inflation daily — famously exposed Argentina's manipulated CPI, (Cavallo & Rigobon, *JEP*, 2016).
- **Mobile phone data:** call density and airtime purchases proxy for consumption in developing countries.

When darkness sheds light...



When darkness sheds light…



*Countries with over 5m people, freedom status in 2021 †In 2021 $ at market exchange rates, assuming reported 1992 GDP figures are accurate

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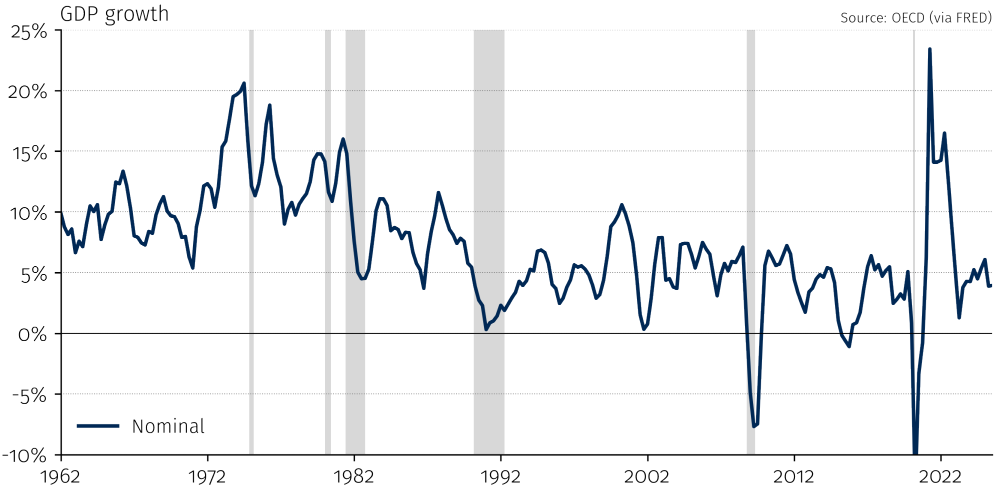
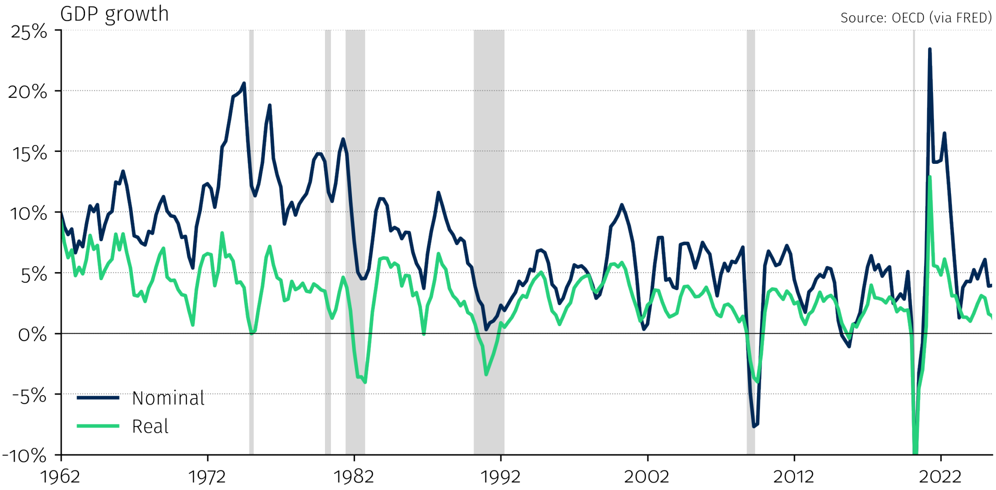
Comparing GDP over time

Nominal GDP growth in Canada



Nominal vs. real GDP growth in Canada



GDP and prices

Definition

- **Nominal GDP** = output valued at *current* prices
- **Real GDP** = output valued at *constant* prices
- **GDP deflator** = $\frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100$

Key insight

Nominal GDP comparisons are contaminated by price changes. Real GDP isolates quantity changes, allowing us to measure true economic growth.

What if higher prices reflect better quality?

When a good/service **improves**, part of the price increase reflects quality, not inflation:

- **New cars:** average price rose from ∼\$23K (2005) to ∼\$48K (2024) — but modern cars have 10+ airbags, lane assist, backup cameras, and last twice as long
- **Cancer drugs:** a year of treatment cost ∼\$10K in 2000; today's immunotherapies exceed \$100K — but the cancer mortality rate has fallen 34% since 1991

Key insight

Statistical agencies try to adjust for this (**hedonic pricing**), but actual quality growth is roughly **3× larger** than what they measure. If we overstate inflation, we understate real GDP growth...

Bils & Klenow, *AER*, 2001

Inflation is back in fashion…

B — "The Affordability Crisis Should Have Been Over By Now"

THE GLOBE AND MAIL — "Canada's inflation rate rises to 2.4% as grocery prices surge"

WSJ — "Trump's Tariffs Cost the Average Household $1,000 Last Year"

wp — "Rising prices fueled by tariffs, delayed inflation report shows"

LE DEVOIR — "Vers une nouvelle hausse du prix des aliments en 2026"

LA PRESSE — "L'inflation alimentaire ne disparaîtra pas de sitôt"

What is inflation?

Inflation

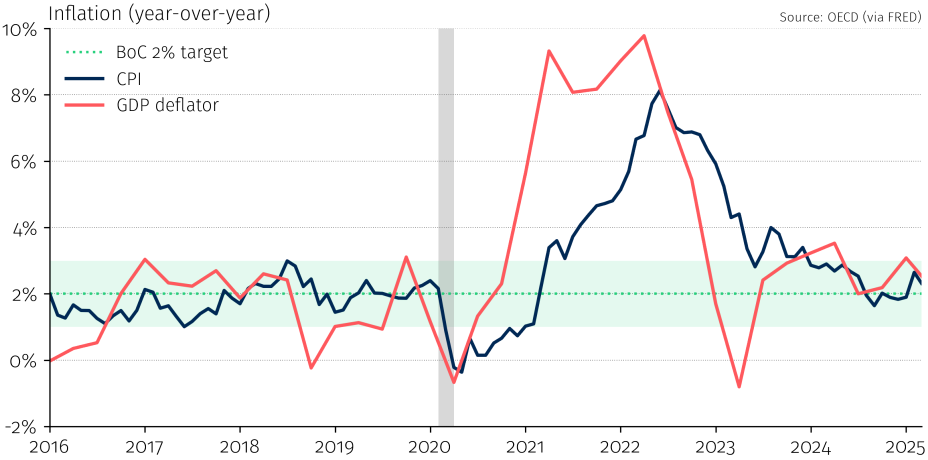
The % rate of change in the **overall price level** between two periods.

$$\pi_t = \frac{P_t - P_{t-1}}{P_{t-1}} = \%\Delta P \quad \text{where } P_t = \text{price level at time } t$$

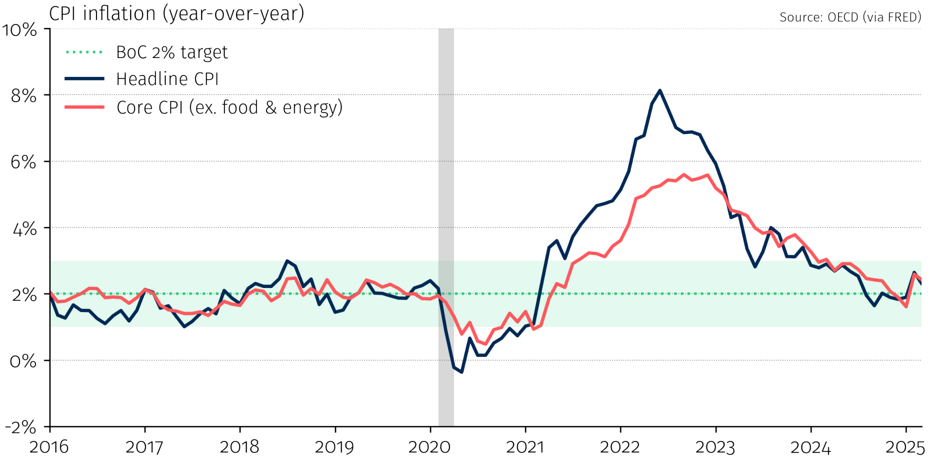
Key question: how do we measure P_t?

- **GDP deflator:** prices of all goods produced domestically (broad)
- **Consumer Price Index (CPI):** prices of a basket of goods bought by consumers (most commonly reported)

Recent inflation in Canada

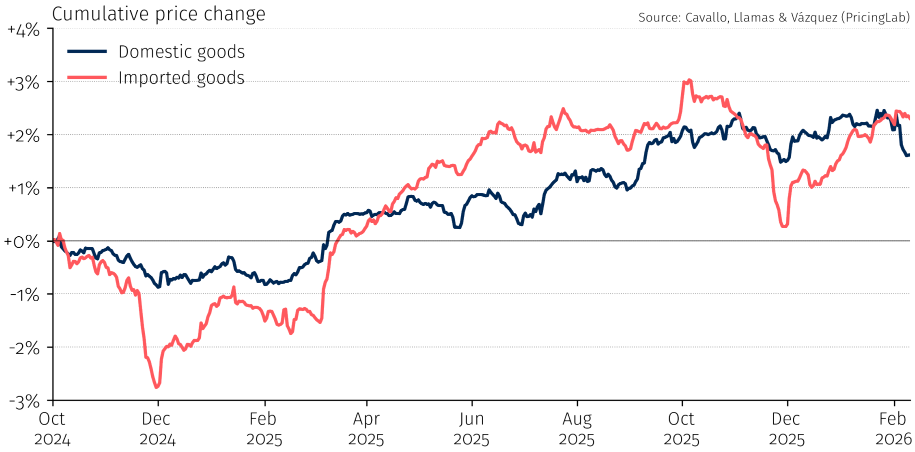


Headline vs. core inflation



High-frequency inflation indicators

Scraping **millions of online prices daily**, for a real-time alternative to official CPI data



New product bias and the price of light

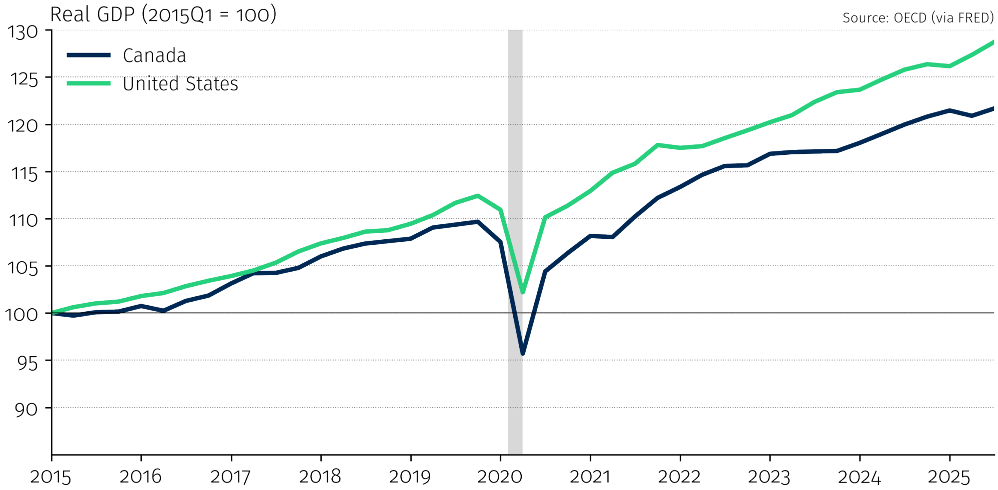
When an entirely **new product** appears, there is no predecessor to compare it to. The price index starts fresh, missing the leap...

The price of light (Nordhaus, 1997)

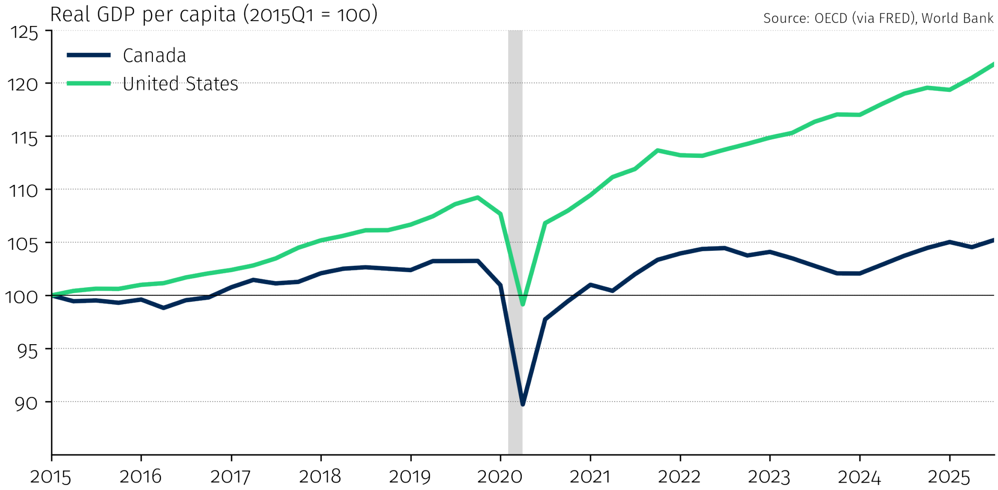
- What are we buying when we buy a candle, a kerosene lamp, or a light bulb?
- We are buying **light** — lumens!
- From candles to LEDs, the true cost per lumen fell by a factor of $\sim$500.
- But price don't track "lumens" — they track **products** (e.g., candles, lamps).
- Each technology switch **resets** the comparison — the revolution is invisible.

Comparing GDP across countries

Canada is keeping up with the United States...



Until you account for population growth...



GDP per capita: enormous variation

[Figure: GDP per capita (PPP) since 1950]
US, Canada, Germany, S. Korea, China, India, Nigeria]
Source: Our World in Data / Penn World Table

Comparing GDPs across countries

Why exchange rates can be misleading

Two ways to convert GDP into a common currency:

1. **Market exchange rates**
   - Useful for comparing *economic power* (e.g., trade, geopolitics)
   - Problem: very volatile, and ignores price-level differences
2. **Purchasing Power Parity (PPP)**
   - Adjusts for differences in the cost of living
   - Better measure of *living standards*
   - A dollar buys much more in Lagos than in New York

China vs. United States: it depends how you measure

[GDP at market exchange rate]
US > China

[GDP at PPP]
China > US since ~2016

Which measure to use?

It depends on the question. Geopolitical influence → market rates. Living standards → PPP.

Beyond GDP

What GDP does not capture

GDP is a useful but incomplete measure of well-being. It misses:

- **Inequality:** a country's GDP can grow while most citizens get poorer
- **Non-market production:** household work, childcare, volunteering
- **Environmental costs:** pollution, resource depletion, climate damage
- **Health and well-being:** life expectancy, mental health, leisure time
- **Quality improvements:** your smartphone is infinitely better than a 2005 phone, but GDP struggles to capture this

Growth has lifted millions out of extreme poverty...

[Figure: Share of population in extreme poverty by region, 1981–2020]
Source: Our World in Data / World Bank

...and saved millions of lives

[Figure: Life expectancy at birth, selected countries, 1543–2020]
Source: Our World in Data (Riley, Clio Infra, UN)

But growth alone is not enough

Growth with rising inequality?

- U.S. real GDP per capita has roughly doubled since 1980
- But median household income has barely budged
- The gains have been concentrated at the top

[Figure: GDP per capita vs. median income (US)]

For managers

Inequality shapes consumer demand, labor markets, regulation, and political risk.

Where are we going next?

Connecting the themes to the course

Theme	Course sessions
Why are some countries rich?	S2: Long-run growth
AI, inequality, labor shortages	S3: Labor market
Where does inflation come from?	S4: Money & business cycles
Rate hikes, stimulus, debt	S5: Monetary & fiscal policy
Trade, exchange rates, China	S6: Financial markets & international

Key takeaways

1. **GDP** measures the total value of final goods and services produced in a country
2. The expenditure identity ($Y = C + I + G + NX$) is the backbone of macro accounting
3. Always use **real GDP per capita** over time and **PPP-adjusted GDP** across countries
4. **Inflation** (measured by CPI) tracks changes in the cost of living — focus on **core** measures for the trend
5. GDP data is imperfect: think critically about the numbers and who produces them